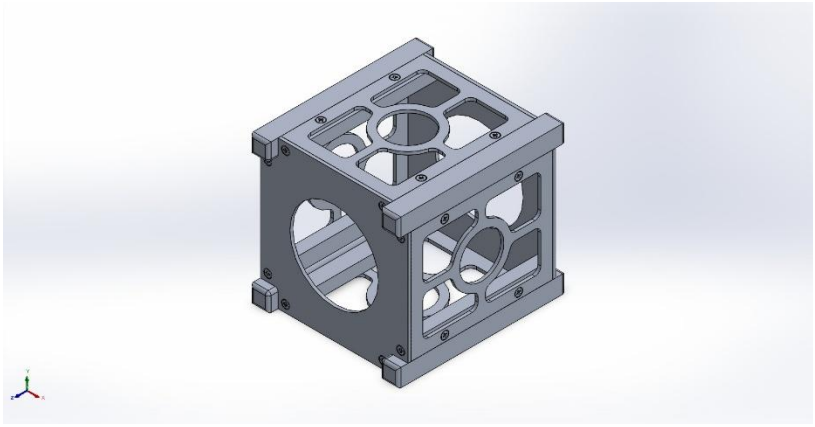


Simulazione di CubeSat

Data: martedì 27 ottobre 2026
Designer: Cristian Caruso
Nome studio: Analisi statica 2
Tipo di analisi: Analisi statica

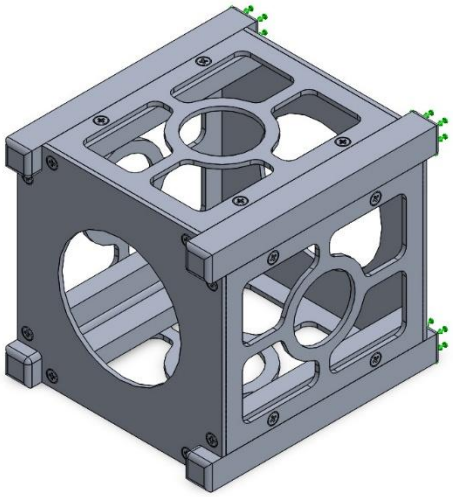
Sommario

Descrizione 1
Informazioni sul modello 2
Proprietà studio 11
Unità 12
Proprietà materiale 13
Carichi e fissaggi 16
Informazioni relative al mesh 17
Risultati studio 19
Conclusione 21

Descrizione

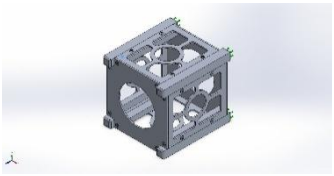
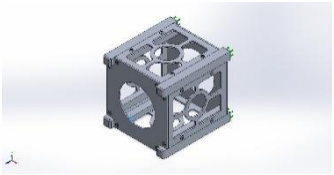
Analisi termo-strutturale di un telaio CubeSat 1U in Alluminio 6061-T6. Lo studio valuta le deformazioni e gli sforzi meccanici indotti dal gradiente termico di un'orbita terrestre (da -40 °C lato eclissi a +80 °C lato Sole) calcolato nella precedente analisi termica, mantenendo la struttura vincolata ai piedini.

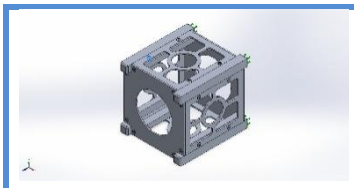
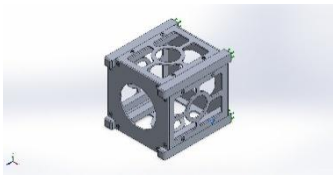
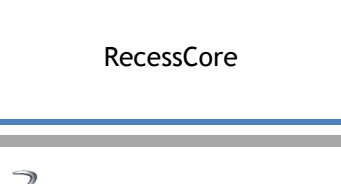
Informazioni sul modello

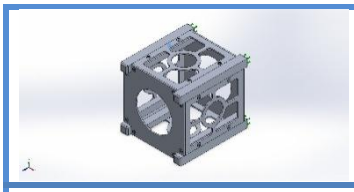
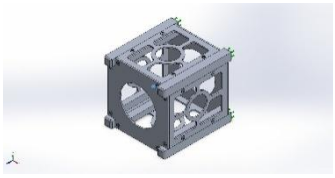


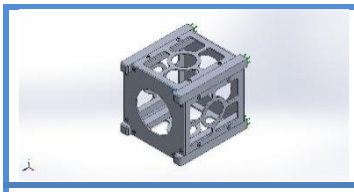
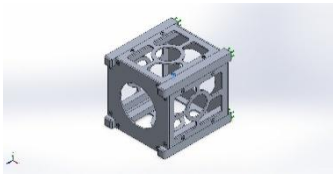
Nome del modello: CubeSat
Configurazione corrente: Default

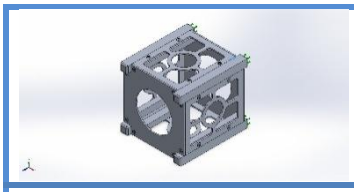
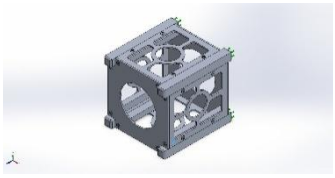
Corpi solidi

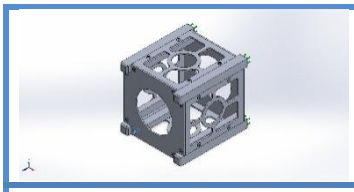
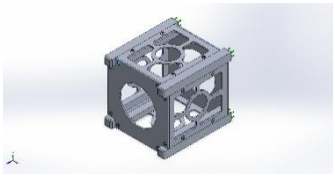
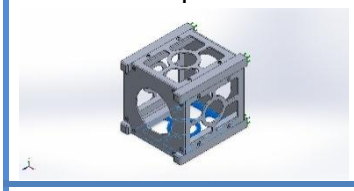
Nome documento e Riferimento	Trattato come	Proprietà volumetriche	Percorso/Data documento modificati
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3**	**C:\SOLIDWORKS Data\browser\ISO\bolts**

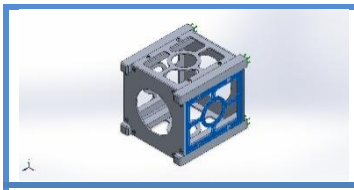
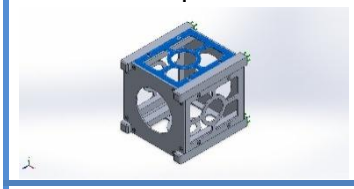
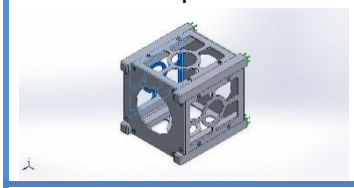
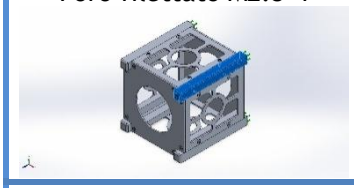
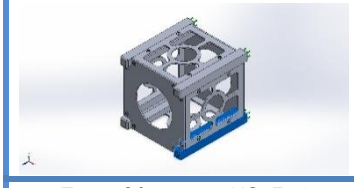
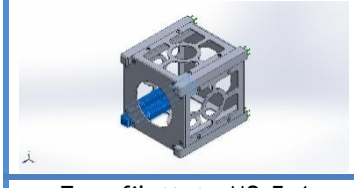
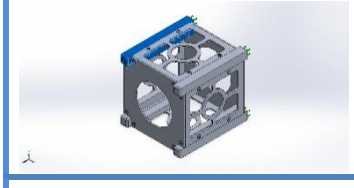
		**Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head**

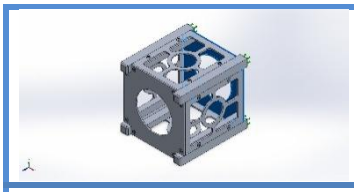
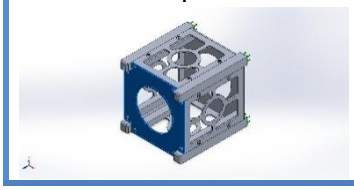
			**screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head**

			**screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head**

			**screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head**

			**screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
RecessCore 	Corpo solido	**Massa:0,000125057 kg Volume:4,63173e-08 m^3 Densità:2.700 kg/m^3 Spessore:0,00122556 N**	**C:\SOLIDWORKS Data\browser\ISO\bolts and screws\cross-recessed head screws\countersunk flat head cross recess screw_iso.sldprt** Oct 27 01:26:29 2026
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro svasato per Vite M2.5-1	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026

			
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0181299 kg Volume:6,71478e-06 m^3 Densità:2.700 kg/m^3 Spessore:0,177673 N**	**C:\Users\carus\Desktop\CubeSat\lateral_panel.SLDPRT** Oct 27 00:37:28 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro filettato M2.5-4 	Corpo solido	**Massa:0,0846295 kg Volume:3,13443e-05 m^3 Densità:2.700 kg/m^3 Spessore:0,829369 N**	**C:\Users\carus\Desktop\CubeSat\rail.SLDPRT** Oct 27 01:08:46 2026
Foro svasato per Vite M2.5-1	Corpo solido	**Massa:0,0351387 kg Volume:1,30143e-05 m^3 Densità:2.700 kg/m^3**	**C:\Users\carus\Desktop\CubeSat\top_down_plate.SLDPRT**

		**Spessore:0,344359 N**	Oct 27 00:12:19 2026
Foro svasato per Vite M2.5-1 	Corpo solido	**Massa:0,0351387 kg** **Volume:1,30143e-05 m^3** **Densità:2.700 kg/m^3** **Spessore:0,344359 N**	**C:\Users\carus\Desktop\CubeSat\top_down_plate.SLDPRT** Oct 27 00:12:19 2026

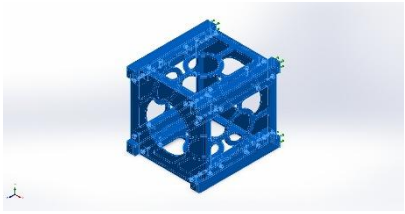
Proprietà studio

Nome studio	Analisi statica 2
Tipo di analisi	Analisi statica
Tipo di mesh	Mesh di elementi solidi
Effetto termico:	Attivo
Opzione termica	Da studio termico
Immetti studio termico:	Termica 1
Frazione di tempo	1
Temperatura di deformazione zero	298 Kelvin
Includi effetti della pressione fluida da SOLIDWORKS Flow Simulation	Disattivo
Tipo di solver	Automatica
Effetto in piano:	Disattivo
Molla morbida:	Disattivo
Sostituzione inerziale:	Disattivo
Opzioni di unione non congruenti	Automatica
Fattore di scala rigidità penalità contatto	1
Grande spostamento	Disattivo
Computa forze del corpo libero	Attivo
Attrito	Disattivo
Usa metodo adattivo:	Disattivo
Cartella Risultato	Documento SOLIDWORKS (C:\Users\carus\Desktop\CubeSat)

Unità

Sistema di unità:	SI (MKS)
Lunghezza/Spostamento	mm
Temperatura	Kelvin
Velocità angolare	Rad/sec
Pressione/Sollecitazione	N/m^2

Proprietà materiale

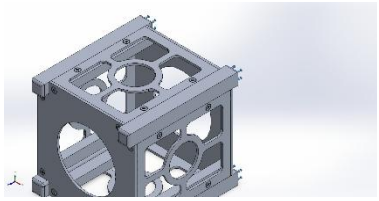
Riferimento modello	Proprietà		Componenti
	**Nome:**	6061-T6 (SS)	CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-1), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-10), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-11), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-12), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-13), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-14), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-15), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-16), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-17), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-18), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-19),
	Tipo di modello:	Isotropico elastico lineare	
	Fallimento di default:	Sollecitazione massima von Mises	
	Snervamento:	2,75e+08 N/m^2	
	Resistenza alla trazione:	3,1e+08 N/m^2	
	Modulo elastico:	6,9e+10 N/m^2	
	Coefficiente di Poisson:	0,33	
	Densità di massa:	2.700 kg/m^3	
	Modulo di taglio:	2,6e+10 N/m^2	
	Coefficiente di espansione termica:	2,4e-05 /Kelvin	

		CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-2), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-20), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-21), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-22), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-23), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-24), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-25), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-26), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-27), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-28), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-29), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-3), CorpoSolido 1(RecessCore)(countersunk

		flat head cross recess screw_iso-30), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-31), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-32), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-4), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-5), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-6), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-7), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-8), CorpoSolido 1(RecessCore)(countersunk flat head cross recess screw_iso-9), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-1), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-2), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-3), CorpoSolido 1(Foro svasato per Vite M2.5-1)(lateral_panel-4), CorpoSolido 1(Foro filettato M2.5-4)(rail-1), CorpoSolido 1(Foro filettato M2.5-4)(rail-2),

		CorpoSolido 1(Foro filettato M2.5-4)(rail-3), CorpoSolido 1(Foro filettato M2.5-4)(rail-4), CorpoSolido 1(Foro svasato per Vite M2.5-1)(top_down_plate-1), CorpoSolido 1(Foro svasato per Vite M2.5-1)(top_down_plate-2)
Dati curva:N/A		

Carichi e fissaggi

Nome fissaggio	Immagine fissaggio	Dettagli fissaggio
Fisso-1		**Entità:** 4 facce **Tipo:** Geometria fissa

Forze risultanti

Componenti	X	Y	Z	Risultante
Forza di reazione(N)	-0,000244141	2,67029e-05	-0,000732422	0,000772502
Momento di reazione(N.m)	0	0	0	0

Informazioni relative al mesh

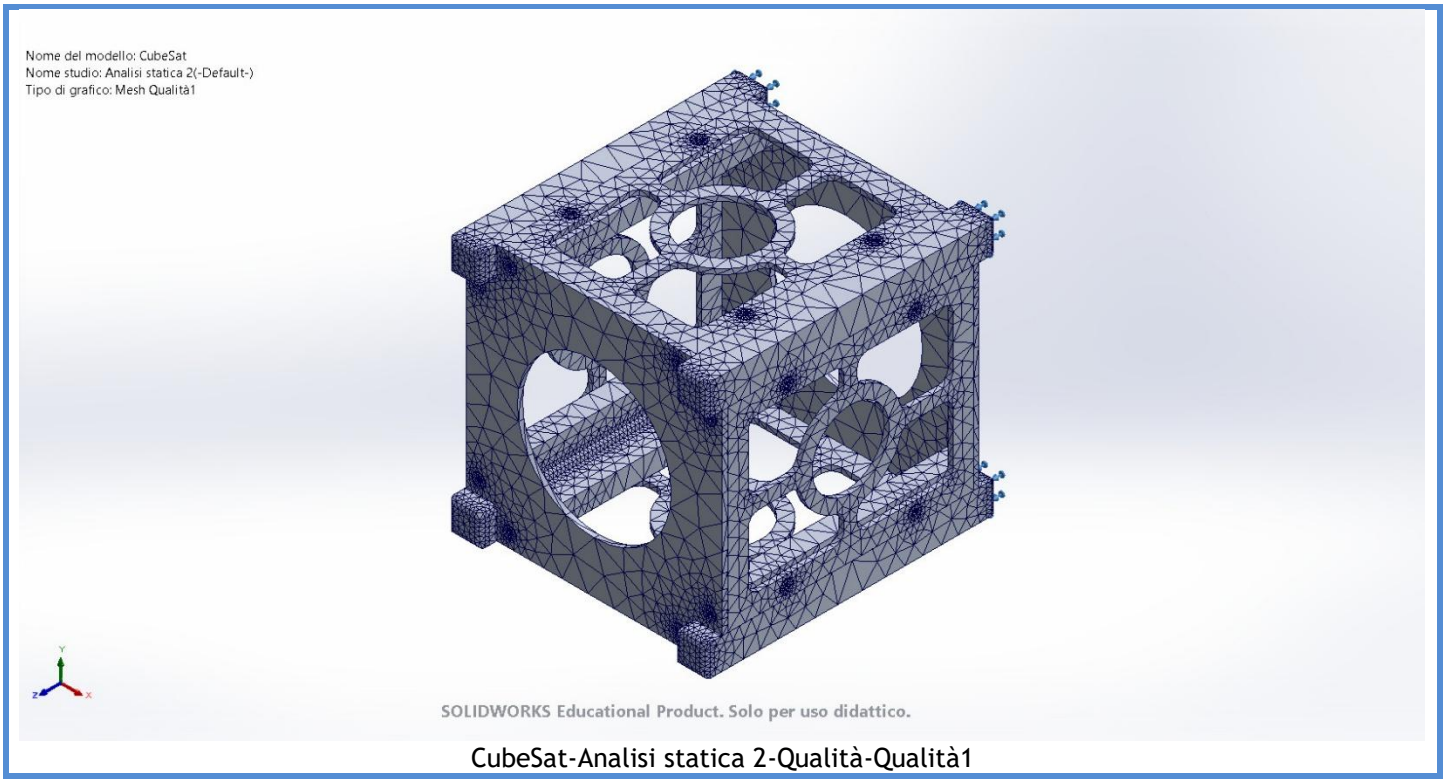
Tipo di mesh	Mesh di elementi solidi
Mesher usato:	Mesh basata su curvatura di blend
Punti jacobiani per mesh di alta qualità	16 Punti
Dimensione massima elemento	13,0766 mm
Dimensione elemento minima	0,653832 mm
Qualità della mesh	Ottima
Ripeti mesh parti non riuscite in modo indipendente	Disattivo
Riutilizza mesh per parti uguali in un assieme (solo mesher basato su curvatura di blend)	Disattivo

Informazioni relative al mesh - Dettagli

Nodi totali	197469
Elementi totali	115468
Rapporto d'aspetto max	15,072
% di elementi con rapporto d'aspetto < 3	91,9
Percentuale di elementi con rapporto d'aspetto > 10	0,0909
Percentuale di elementi distorti	0
Tempo per completare la mesh(hh;mm;ss):	00:00:10
Nome del computer:	

Grafici qualità della mesh

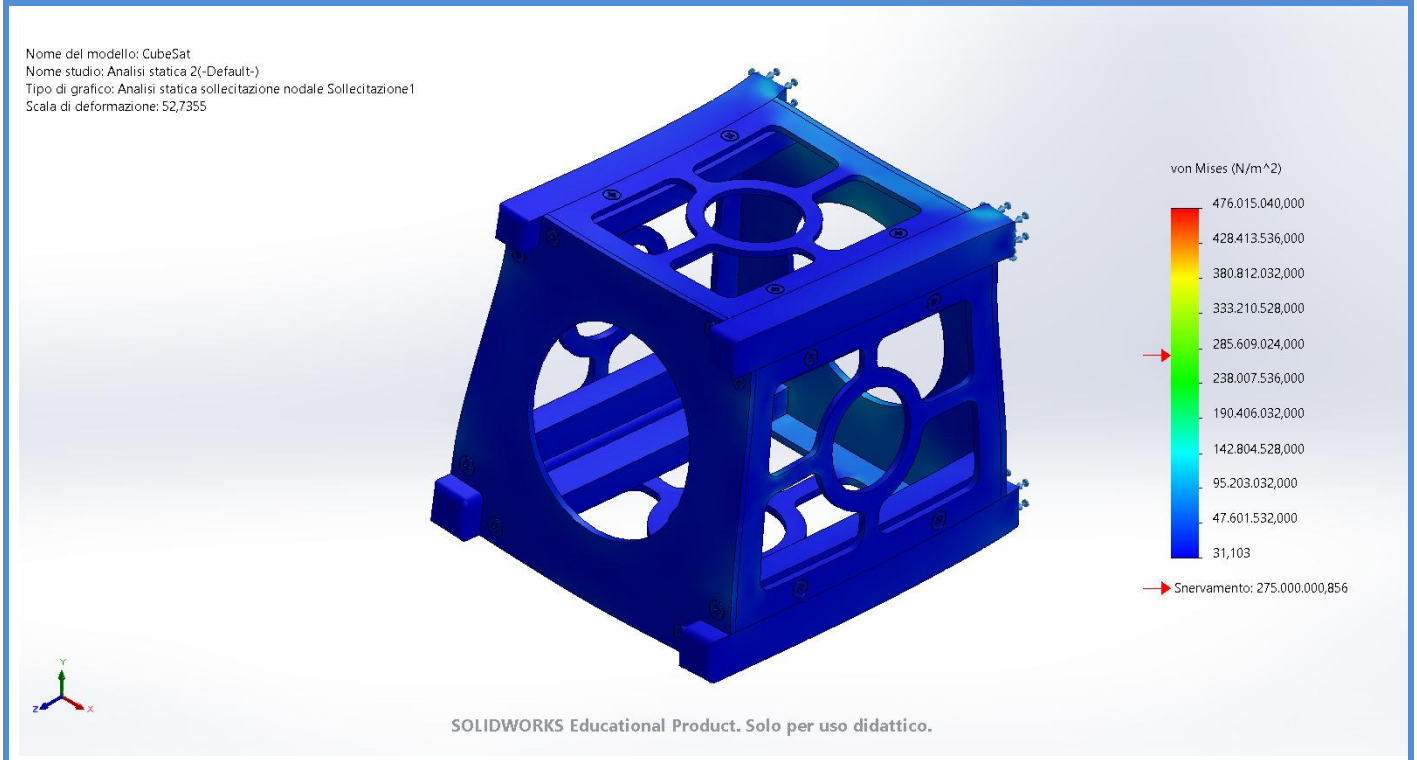
Nome	Tipo	Min	Max
Qualità1	Mesh	-	-



CubeSat-Analisi statica 2-Qualità-Qualità1

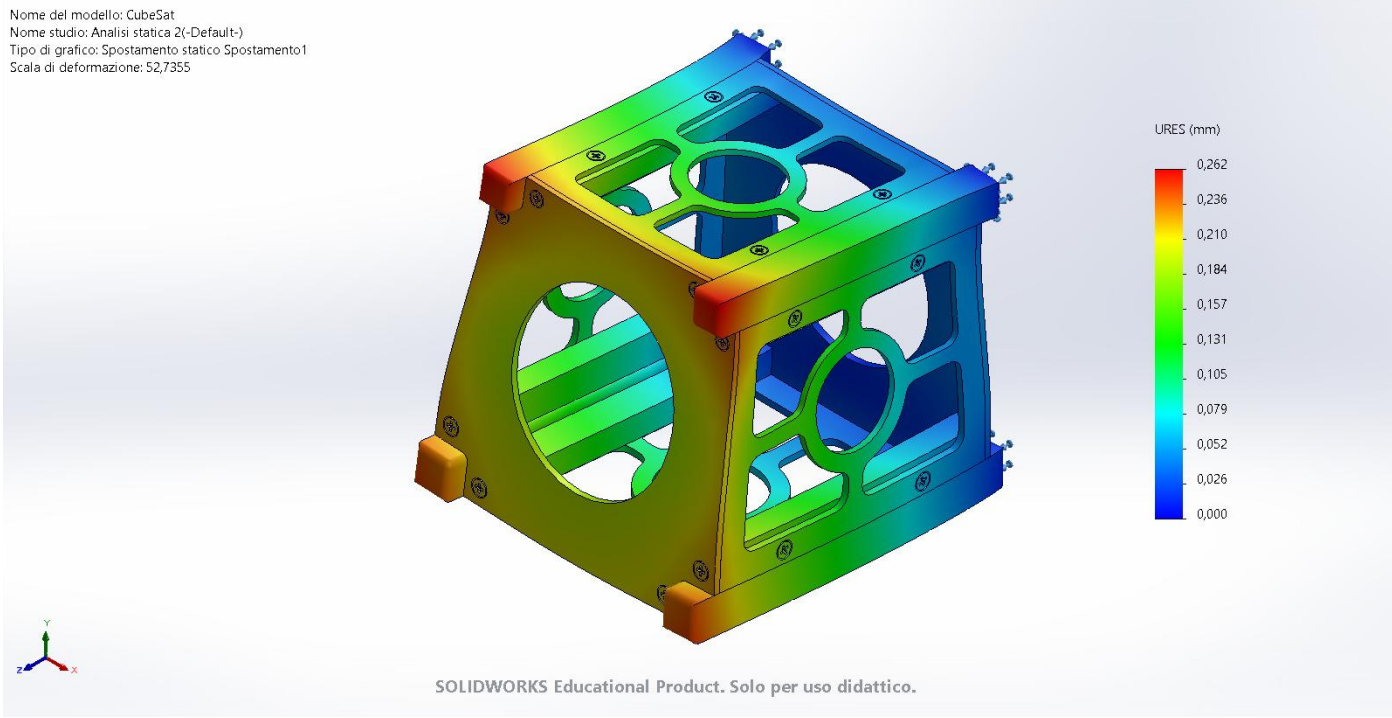
Risultati studio

Nome	Tipo	Min	Max
Sollecitazione1	VON: sollecitazione von Mises	31,103N/m^2 Nodo: 33947	476.015.040,000N/m^2 Nodo: 155336



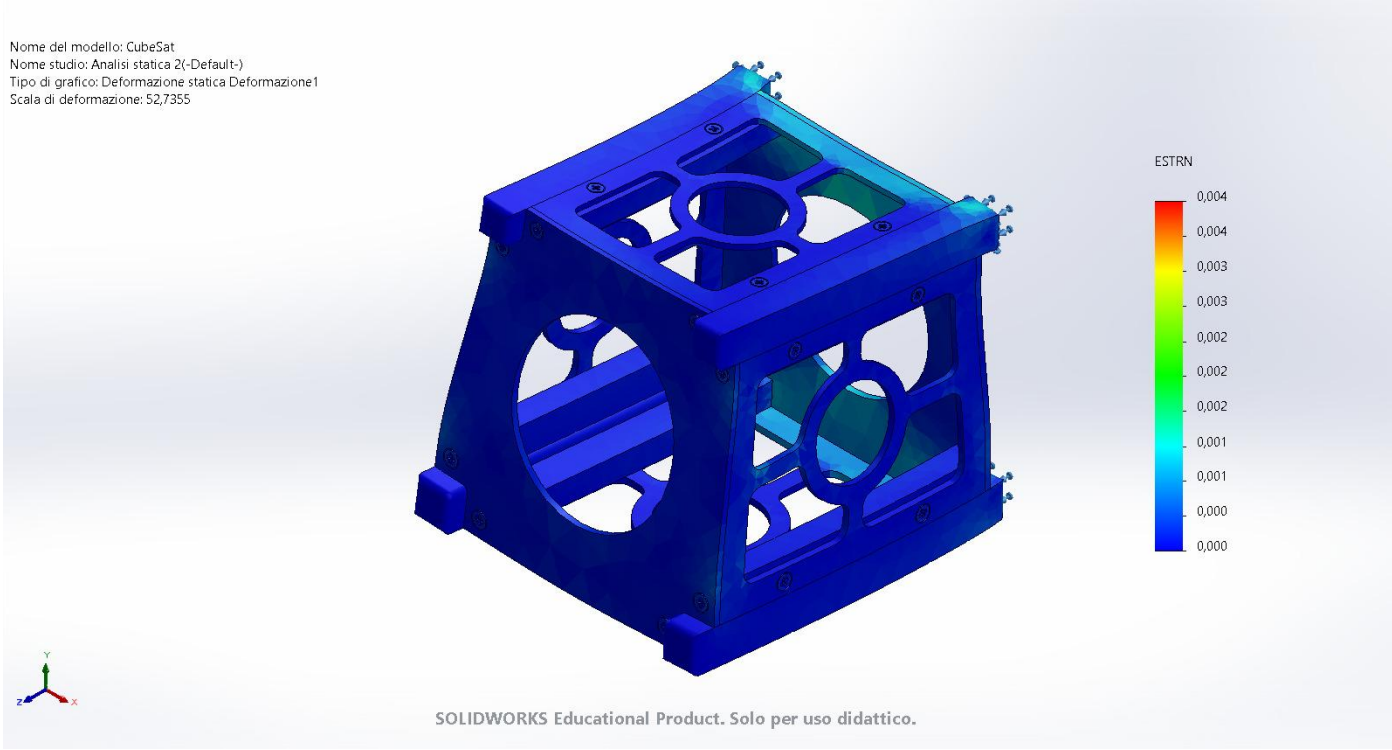
CubeSat-Analisi statica 2-Sollecitazione-Sollecitazione1

Nome	Tipo	Min	Max
Spostamento1	URES: spostamento risultante	0,000mm Nodo: 67439	0,262mm Nodo: 68260



CubeSat-Analisi statica 2-Spostamento-Spostamento1

Nome	Tipo	Min	Max
Deformazione1	ESTRN: deformazione equivalente	0,000 Elemento: 18196	0,004 Elemento: 104331



CubeSat-Analisi statica 2-Deformazione-Deformazione1

Conclusione

L'analisi certifica l'eccellente stabilità dimensionale del telaio nello spazio. La massima dilatazione termica registrata è infinitesimale (0,262 mm), garantendo l'assoluta integrità degli alloggiamenti per le schede elettroniche interne. Il picco tensionale evidenziato dal grafico di Von Mises (476 MPa) rappresenta esclusivamente una singolarità numerica dovuta all'idealizzazione matematica dei vincoli (geometria fissa che impedisce artificialmente l'espansione dei piedini). Nell'effettivo ambiente operativo, le tolleranze meccaniche del dispenser del razzo permettono la naturale micro-dilatazione del materiale, annullando tale stress localizzato. L'intero corpo portante del satellite lavora a tensioni minime, ampiamente al di sotto del limite di snervamento di 275 MPa.